

query model

barrier

Cut queries (Fourier basis)

hyperedges crossing $(S, \bar{S})$

basis mismatch: $\geq 2^d$ queries

a size- d hyperedge $\rightarrow 2^d$ Fourier terms

Edge-detection queries

is there a hyperedge inside S ? (yes/no)

too little information: $(s/d)^{\Omega(d)}$

[Angluin–Chen '08]

Additive queries (*this talk*)

hyperedges inside S

best prior: $O(sn)$ — ignores degree d

or $O(sd \log n)$ under independence assumptions

This work: $O(sd \log(n/d))$ queries — deterministic, near-optimal, no independence assumptions